KapitWatt

Full plan for the IDOL Hackathon 2026 build and demo

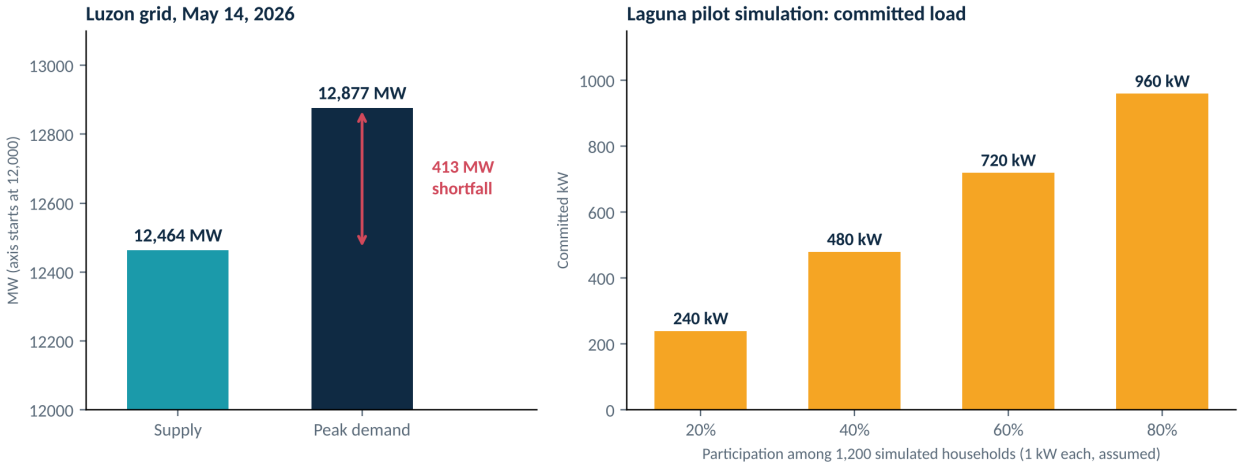
Updated November 12, 2026

KapitWatt is a residential demand response platform. When NGCP raises a yellow or red alert, Meralco's operator alerts opted-in households. Each household commits to cutting a set number of kW during the peak window and is paid for the kWh Meralco can verify it avoided, from smart-meter data or from meter photos taken in the app. Meralco sees committed and delivered load by area in real time. It is a household version of the Interruptible Load Program (ILP).

Scope: Laguna, simulated. The prototype covers seven Meralco franchise towns in Laguna (Santa Cruz, Pila, Victoria, Nagcarlan, Liliw, Majayjay, Magdalena), with 21 sampled barangays and about 1,200 simulated households. Team phones join as real households in Santa Cruz. The software is real; the households are simulated.

5	2	7	24 h
core steps	**apps**	**Laguna pilot towns**	**build window**
alert, commit, shift, measure, reward	web dashboard and mobile app	Santa Cruz plus six neighbors	alert to live dashboard

Why it matters



Left: NGCP's May 14, 2026 red alert, with supply of 12,464 MW against a 12,877 MW peak (proposal sources). Right: committed load in the Laguna simulation at different participation rates, assuming 1 kW per participating household. The right chart is a parameter sweep, not a forecast.

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1. Apps and features

App	Users	Tech	Purpose
Household app	Households in the pilot	Expo + React Native (Android APK)	Receive alerts, commit a cut, take meter photos, earn rewards
Utility dashboard	Meralco operator	Expo web (Expo Router)	Send alerts, watch committed and delivered load
API	Both apps and simulator	Hono on Cloudflare Workers	Alerts, commitments, readings and checks, rewards, push dispatch
Laguna simulator	Team, demo only	TypeScript package + API routes	Seeds about 1,200 households and runs simulated events
Shared package	All of the above	TypeScript, Zod	Types, schemas, baseline, payout and check rules

Priority: P0 must work in the demo. P1 if time allows. P2 shown as a slide or stub.

Household app

Feature	What it does	Priority
Sign-up and opt-in	Pick town and barangay, consent to alerts and data use (Data Privacy Act 2012), register meter type and serial number.	P0
Commit a cut	Choose how many kW to cut (presets of 0.5, 1, 1.5 and 2 kW, or a stepper). The maximum depends on the household's baseline and trust score.	P0
Rehearsal event	Photo households only: a practice window with the same two photos at sign-up. It doubles as the tutorial and sets the baseline.	P0
Meter photo capture	In-app camera only, no gallery. The household types the reading and attaches the photo. Before photo within 15 min of window start, after photo within 30 min of window end, no resubmission.	P0
Grid alert	Push notification for a yellow or red alert with the peak window. SMS is mocked.	P0
Event screen	Countdown, commitment, photo deadlines and a barangay progress bar. Smart-meter households also see their live reduction against baseline.	P0
Result card	After settlement: verified cut, bill credit, barangay bonus and points.	P0
Points, leaderboard, history	Points per verified kWh, barangay ranking and past events.	P0
Appliance helper	Optional list with rated kW that suggests what to switch off to reach the cut. Not used for payment.	P1
Trust score and commit cap	Rises with clean settled events, falls with flagged readings, and sets the largest cut a household can commit.	P1
Home monitor link, bill credit redemption	Roadmap slides only.	P2

Utility dashboard

Feature	What it does	Priority
Alert form	Level, window, target barangays, target kW, message preview and a confirm step. Sends the push.	P0
Live committed load	Committed kW updates as households commit; total against the target.	P0
Delivered load by town and barangay	Verified delivered kW against each barangay's target, with a target-hit marker. Drills down from town to barangay.	P0
Measure (fast-forward)	Ends the window, compresses the photo timers in demo mode and settles results so the demo does not wait.	P0

Feature	What it does	Priority
Simulation controls	Participation, follow-through, path mix, photo completion, preset, run event.	P0
Flagged readings list	Photos that failed a rule check, with the readings and flags. A person approves or rejects each one.	P1
Schedule, reminder, cancel, alert log	Schedule a send, reminder push before the window, end an alert, record who sent what.	P1
Load curve and relief forecast	Baseline versus actual with relief shaded; committed times follow-through rate.	P1
Feeder-level check	Compare barangay feeder load with forecast to confirm delivery.	P2

API and shared package

Feature	What it does	Priority
Auth and opt-in	Simple token auth for the demo; stores push tokens and meter registration.	P0
Alert and commitment endpoints	Create alerts, record commitments (cut kW and path), return live totals.	P0
Meter reading endpoints	Accept before, after and rehearsal readings with the photo; run the rule checks; flag failures.	P0
Push dispatch	Sends to opted-in tokens in targeted barangays through Expo Push.	P0
Baseline and kWh avoided	Baseline from the rehearsal or AMI history; verified kWh capped by the commitment and bill history.	P0
Settlement	At window end plus 30 min: results, barangay target check, credit, points and bonus.	P0
Simulator routes	/sim/seed, /sim/run, /sim/reset, /sim/status, protected by a simulator key.	P0
AMI feed adapter	Reads Meralco interval data. Simulated in the demo; production needs a Meralco data agreement and customer consent.	P1

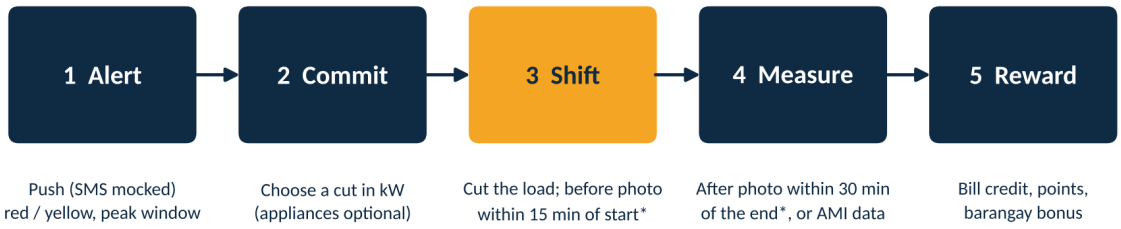
Alert form settings

Setting	Detail	Priority
Level	Yellow or red. Red can ask for a bigger cut.	P0
Window	Start and end, default 2 hours, with quick presets.	P0
Target barangays	Pick towns or barangays, or all pilot areas.	P0
Target	kW or MW of relief wanted; the dashboard compares against it.	P0
Message preview and confirm	Default text the operator can see. Confirm: "Send red alert to 7 towns, about 1,200 households?"	P0
Send now or schedule, reminder, cancel	Scheduled send, a reminder 15 minutes before, and an alert-lifted push.	P1

Guardrails: only opted-in households get alerts, and every alert is logged with sender and time.

2. User flows

Household flow

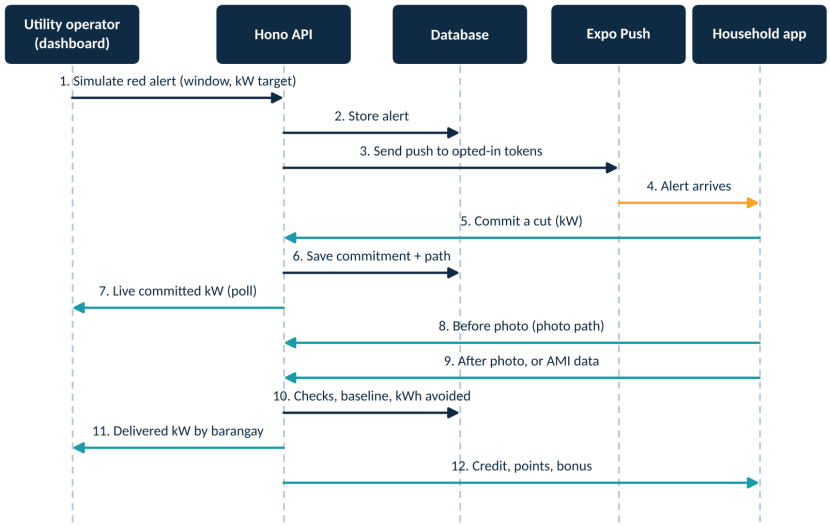


** Photo households only. Smart-meter households need no photos; Meralco's interval data is used.*

Once at sign-up: opt in, register meter type and serial, do a rehearsal event (photo households). Steps 1 to 5 repeat for every alert.

Step	What the household sees	What the system does
0 Sign-up (once)	Opt in, register meter type and serial. Photo households do a rehearsal event.	Stores the registration and sets the baseline.
1 Alert	Push: red alert, peak window, ask to cut load.	Stores the alert and sends to opted-in tokens.
2 Commit	Choose a cut in kW. The appliance list is an optional helper.	Caps the cut by baseline and trust score; saves commitment and path.
3 Shift	Window opens. Photo households take the before photo within 15 min.	Runs the checks; the barangay progress bar updates.
4 Measure	Photo households take the after photo within 30 min of the end. Smart-meter households do nothing.	Computes avoided kWh from the readings or the AMI feed.
5 Reward	Result card with credit, points and barangay bonus.	Updates the leaderboard and delivered load.

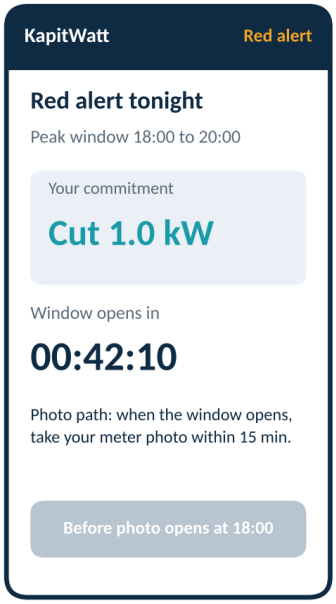
Demo loop: who talks to whom



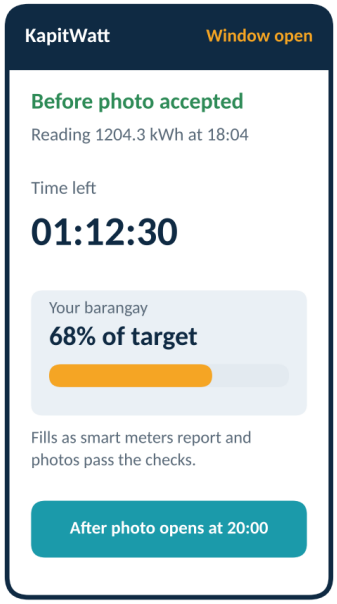
The operator raises an alert, the app receives it and households commit. Photo households send before and after photos; smart-meter households send nothing. The API checks, settles and returns rewards.

What the household sees during an event

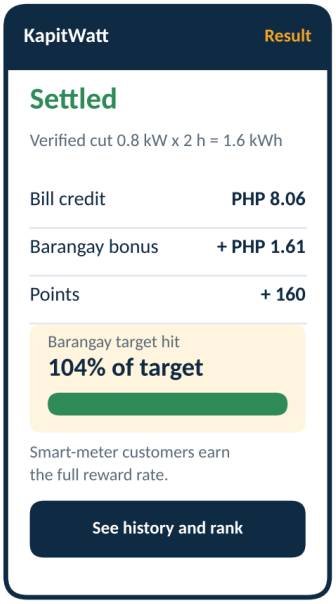
Photo-household view. Smart-meter households skip both photos and see their live reduction instead.



1 Committed, before the window



2 Window open

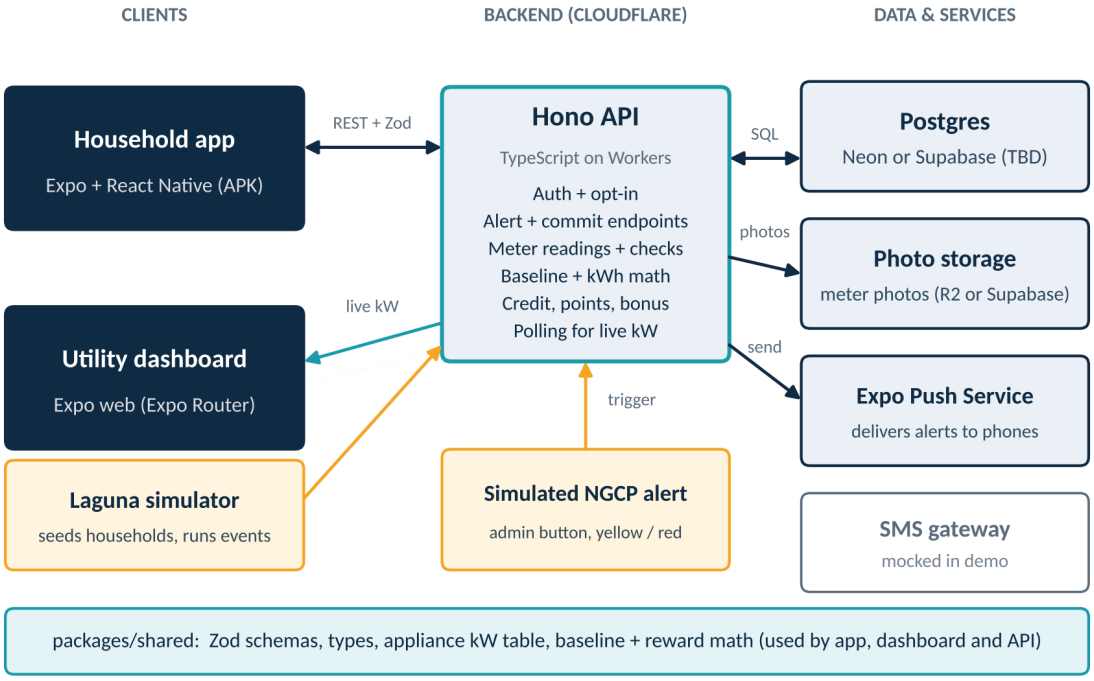


3 Settled

Element	How it works
Commitment	The household commits to a number of kW to cut, not to a list of appliances. Nobody has to prove which device was switched off.
Barangay progress bar	Verified delivered kW in the household's barangay against that barangay's target. It fills from smart-meter data and from photos that pass the checks, so for photo households part of it fills only after settlement.
Barangay bonus	Individuals are paid on their own verified cut. If the barangay reaches 100% of its target at settlement, every household there with a verified cut gets a bonus on its own credit. Households without a verified cut get nothing from it.
Smart-meter households	No photos. They see their live reduction against baseline during the window.

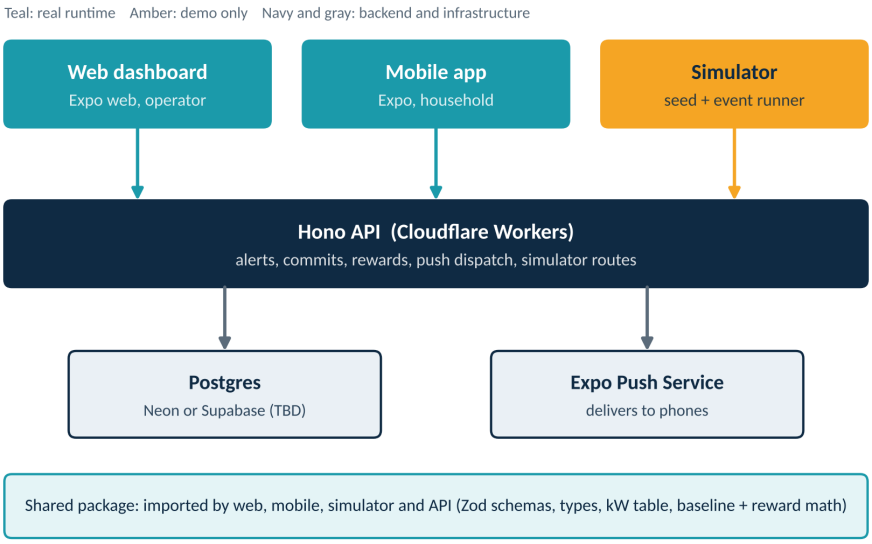
3. Architecture and stack

Layer	Tool	Used for
Language	TypeScript (strict)	All code: app, dashboard, API, simulator, shared package.
Monorepo	pnpm workspaces	apps/mobile, apps/dashboard, apps/api, packages/shared, db, sim, config.
Mobile	Expo + React Native, Expo Router	Household app. APK via EAS; Expo Go during development.
Dashboard	Expo web (Expo Router)	Utility view sharing components and types with the app.
Push	expo-notifications, Expo Push Service	Alert delivery. Needs real phones to test.
API	Hono on Cloudflare Workers	REST, alert trigger, baseline and reward logic.
Validation	Zod (in packages/shared)	One schema used everywhere.
Client data	TanStack Query	Fetching and refetching live totals.
Database	Postgres on Neon or Supabase (undecided)	Households, alerts, commitments, meter readings, rewards.
Photo storage	Cloudflare R2 or Supabase Storage	Meter photos in a private bucket.
Live updates	Short polling (about 2 seconds)	Dashboard totals. SSE or Supabase Realtime only if time allows.



Scope: Laguna pilot simulation. NGCP alerts, households and reward rates are simulated or placeholders.

4. Monorepo and connections



All three clients call the same API. The simulator is an amber, demo-only client. The shared package is imported by everything, so commit, baseline and reward logic is identical for phones and simulated households.

Monorepo layout

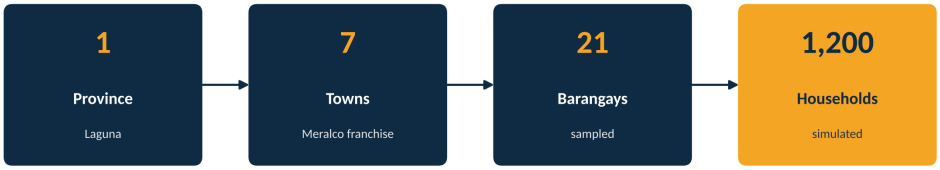
```
kapitwatt/
  apps/
    mobile/       Expo + React Native, household app (APK via EAS)
    dashboard/    Expo web, utility dashboard
    api/          Hono on Cloudflare Workers
  packages/
    shared/       Zod schemas, types, kW table, baseline + reward math
    db/           Postgres schema, migrations, query helpers
    sim/          Laguna seed data + event runner
    config/       shared tsconfig and lint settings
  pnpm-workspace.yaml
  .npmrc          node-linker=hoisted
```

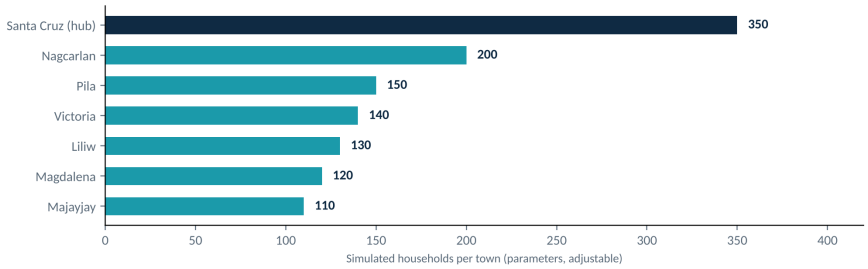
Rules and setup

Rule	Why
Apps never import each other; they talk through the API and shared package.	Keeps work parallel and merge conflicts low.
The simulator calls the API and shared functions, never inserts rows directly.	The demo then exercises the real code paths.
Only apps/api and packages/db touch Postgres.	The Neon or Supabase choice stays in one package.
Push is sent only from the API, through Expo Push.	One place for opt-in and targeting checks.
Export TypeScript source from internal packages; use workspace:* deps.	No build step during the 24 hours.
Hoisted node-linker; one pinned react and react-native version at the root.	Avoids Metro symlink and duplicate React problems.
One env var, EXPO_PUBLIC_API_URL, for both Expo apps.	Same config for mobile and web.
Deploy: API with wrangler to Workers; dashboard via expo export on Cloudflare Pages; mobile via EAS as an APK.	Same three targets for the whole team.

5. Laguna simulation and simulator

The pilot covers Meralco franchise towns in Laguna. Households, appliance declarations, commitments and delivered kWh are simulated. The software, the alert flow, the baseline math and the dashboard are real.





Household counts per town are simulation parameters. Barangay names are pulled from the PSA list at seed time.

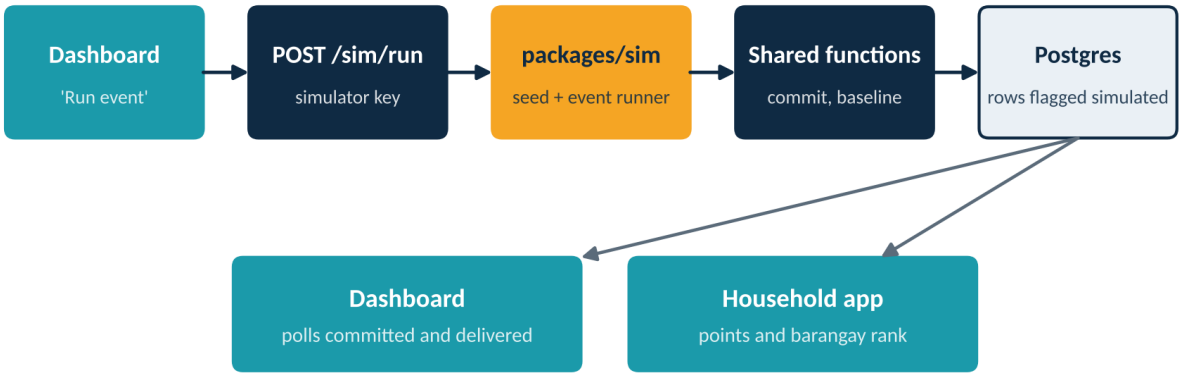
Simulation parameters

Parameter	Default setting	Note
Towns	Santa Cruz (hub), Pila, Victoria, Nagcarlan, Liliw, Majayjay, Magdalena	Meralco franchise towns in Laguna.
Barangays	21 sampled across the 7 towns	Names from the PSA list.
Households	1,200 simulated, plus team phones as real households in Santa Cruz	Real phones prove push and the app.
Cut size	1 kW committed per participating household	Aircon figure from the proposal; assumption.
Participation	60%, adjustable 20% to 80%	Assumption.
Follow-through	80% of committed kW delivered	Assumption, adjustable per path.
Verification path mix	15% smart meter, 85% photo	Assumption, not real Laguna meter data.
Photo completion	90% of photo households submit both photos on time and pass the checks	Assumption.
Baseline noise	Plus or minus 10% around each household's hourly baseline	Makes the load curve look real.
Alerts	Yellow or red, about 2 hours, targeted by barangay	May 14 brownouts ran about two hours each, listed by barangay.

Simulator features

Feature	What it does	Priority
Seed Laguna	Creates 7 towns, 21 barangays, about 1,200 households and appliances.	P0
Run event	For targeted barangays, simulated households commit; after the window it generates delivered kWh.	P0
Parameters	Participation, follow-through, path mix, photo completion, baseline noise.	P0
Fake meter feed	Generates AMI interval readings for smart-meter households and before and after readings for photo households, flagged as simulated.	P0
Paced commits	Commits spread over 20 to 30 seconds so the live total rises gradually.	P0
Reset	Clears simulated data only.	P0
Safety	Rows flagged is_simulated, simulator key required, routes off outside the demo environment.	P0
Presets, seeded randomness, run summary	Poor 30%, typical 60%, best 80%; repeatable runs; per-town summary; a seed where the presenter's barangay reaches its target to show the bonus.	P1
Failure injection, real barangay import	No-shows and late commits; names loaded from PSA.	P2

How the simulator connects to the app



The simulator never writes rows directly. It calls the same functions the real app uses.

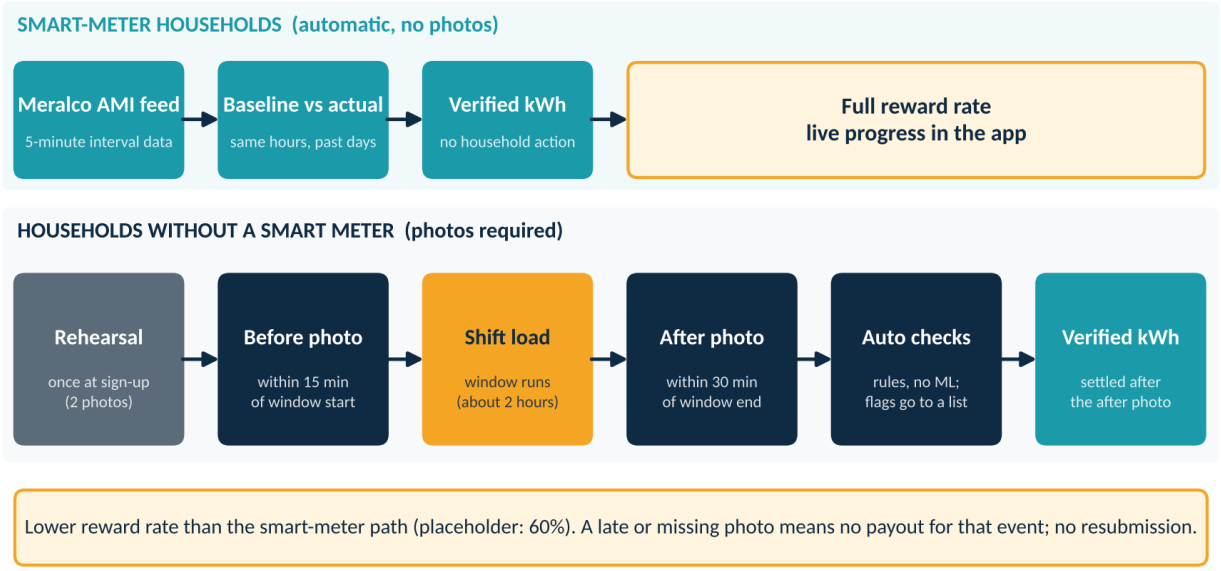
Endpoint	Purpose
POST /sim/seed	Create Laguna households and appliances
POST /sim/run	Run an event for an alert with participation, follow-through, path mix, photo completion and pace
POST /sim/reset	Delete simulated data
GET /sim/status	Counts, last run and current settings

Reads	Writes
The alert (level, window, target barangays); simulated households and appliances; simulation parameters.	Commitments (cut kW, path); meter readings; verified kWh, credit, points and bonus through the reward logic.

The simulator is a backend data generator that runs real logic. The dashboard charts are just the dashboard reading that data. It acts as a service account for simulated rows only, never as a general superadmin.

6. Verification, rewards and data model

The system cannot see which appliance a household switched off. It measures whether consumption dropped and pays for the measured drop. Every payout rests on a meter reading: Meralco's own data, or a photo of the meter taken inside the app.

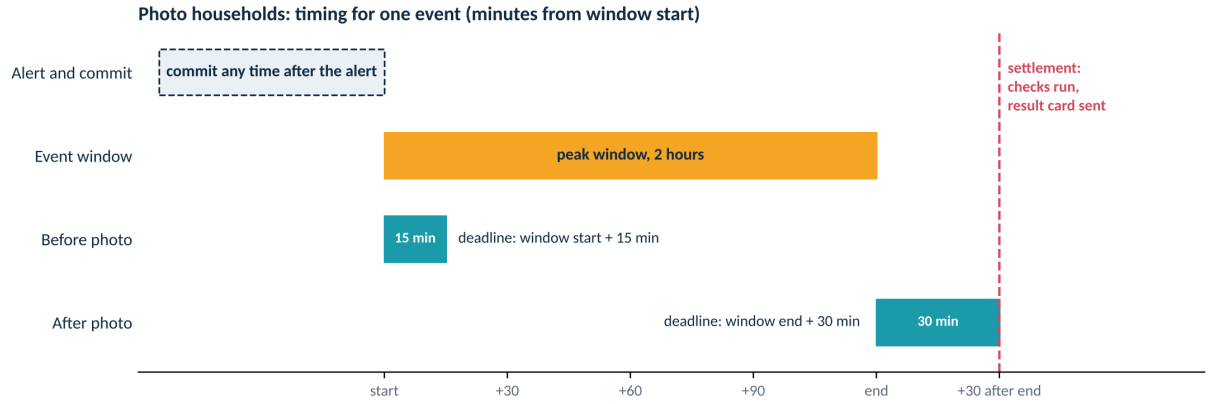


Smart-meter households are verified automatically. Everyone else must take the photos.

	Smart meter	Photo (no smart meter)
Evidence	Meralco 5-minute interval data compared with the household's own history.	Before and after meter photos taken in the app, plus the typed reading.
Household effort	None.	Rehearsal at sign-up, then 2 photos per event.
Reward rate	Full rate.	Lower rate (placeholder: 60%).
Progress shown	Live reduction against baseline.	Result after settlement, at window end plus 30 min.
Failure	Missing data: no payout, flagged.	Late or missing photo, or failed check: no payout, no resubmission.

What we know about smart meters. Meralco had deployed about 196,000 smart meters as of February 2026, against roughly 8 million customers. It plans 72,000 more in 2026 and 3.27 million between 2025 and 2029, and its AMI meters record in 5-minute intervals. We found no route for a household to buy its own billing meter: the meter is Meralco's, so the smart-meter share grows only as the rollout reaches each home. Coverage in the Laguna towns is unknown, so confirm with Meralco and plan for most households to use the photo path. In production the meter path needs a Meralco data feed and customer consent. In the demo it is simulated.

Photo rules



The windows are fixed. A late or missing photo means no payout for that event, and there is no resubmission.

Check	Rule
Capture	In-app camera only; no gallery upload. The server timestamps the upload.
Timing	Before photo within 15 min of window start. After photo within 30 min of window end.
Meter match	The serial is registered at sign-up and must appear in the photo. This also separates meters in a shared bank.
Location	GPS within a set distance of the registered address.
Reading	The household types the reading. It must be higher than the previous one and not more than the service could draw in that time.
Duplicates	A file hash rejects reused photos.
OCR (optional)	An off-the-shelf OCR or vision API reads the display and flags a mismatch with the typed reading. It only flags for review and never rejects on its own. Nothing needs training.
Failures	Rule failures go to the flagged list on the dashboard, where a person approves or rejects them.

All of these are plain rules, so the hackathon build needs no trained model or dataset.

Baseline and payout

A before and after pair only shows what the household used during the event. The baseline says what it would have used anyway.

Path	Baseline	Source
Smart meter	Average kW over the same hours on comparable past days.	AMI history
Photo	kW measured in the rehearsal window at sign-up, with a sanity cap from the bill.	Rehearsal photos and bill history

```
actual_kw     = (reading_after - reading_before) / hours between the two readings
cut_kw       = max(0, baseline_kw - actual_kw)
paid_cut_kw = min(cut_kw, committed_kw, cap_kw)
cap_kw       = 2 x (bill_avg_kwh_month / 720)  (peak factor 2 is a placeholder)
verified_kwh = paid_cut_kw x window hours
credit_php   = verified_kwh x 8.40 x path multiplier  (smart meter 1.0, photo 0.6)
bonus_php    = 20% of credit_php if the barangay reached its target
points        = 100 x verified_kwh
```

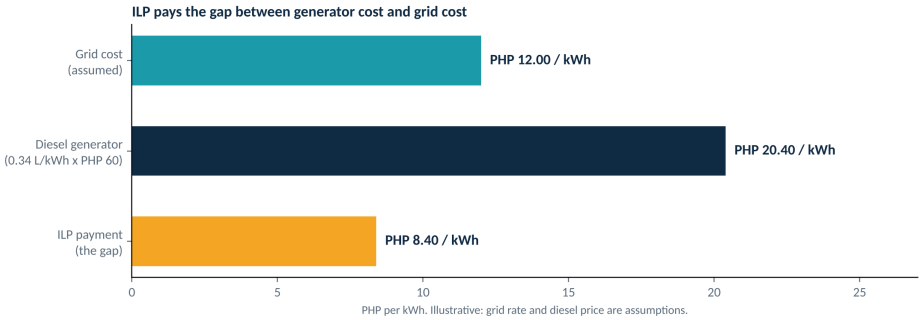
Worked example: one photo household

Step	Value
Event	Window 18:00 to 20:00 (2 h). Committed cut: 1.0 kW.
Readings	Before: 1204.3 kWh at 18:04. After: 1206.6 kWh at 20:10.
Measured load	2.3 kWh over 2.1 h = 1.1 kW.
Baseline	1.9 kW from the rehearsal, so the measured cut is 0.8 kW.
Cap	min(0.8, committed 1.0, bill cap 1.25 kW for 450 kWh a month) = 0.8 kW.
Verified	0.8 kW x 2 h = 1.6 kWh.
Payout	1.6 x 8.40 x 0.6 = PHP 8.06 credit, plus PHP 1.61 barangay bonus, plus 160 points. A smart-meter household with the same cut would get PHP 13.44 credit.

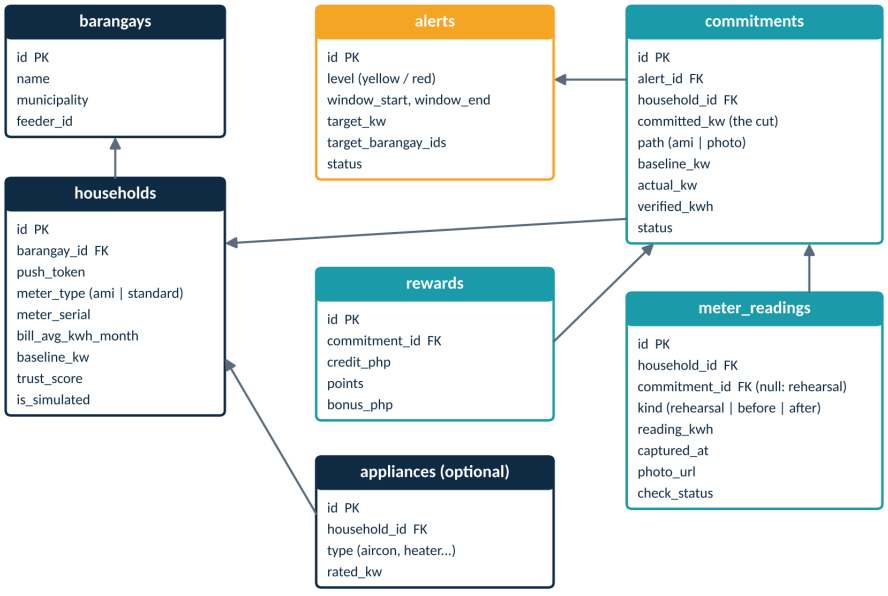
The reward rate and multipliers are placeholders for tuning. A time-aligned elapsed period is used for the measurement, but pay is for the scheduled window, so a late after photo lowers the measured cut slightly instead of adding paid hours.

Rewards, barangay bonus and trust

Rule	Detail
Reward rate	The ILP pays the gap between generator cost and grid cost. Households have no generator, so KapitWatt uses a flat rate per verified kWh. The placeholder is PHP 8.40, the illustrative gap below. The real rate would be set by Meralco and the ERC.
Barangay target	The alert's target kW is split across the targeted barangays by opted-in households. The operator can override a barangay's share.
Progress and bonus	Progress is verified delivered kW divided by the barangay target. At 100% at settlement, each household with a verified cut there gets 20% on top of its own credit (placeholder). No verified cut, no bonus.
Delivery ratio	An approved photo cut counts at full weight toward the barangay total.
Trust score (P1)	Starts neutral, rises with clean settled events and falls with flagged readings. It sets the largest cut a household can commit and how often it is audited.



Data model



Arrows run from the table holding the foreign key to the table it references. appliances is an optional estimator; nothing is paid from it.

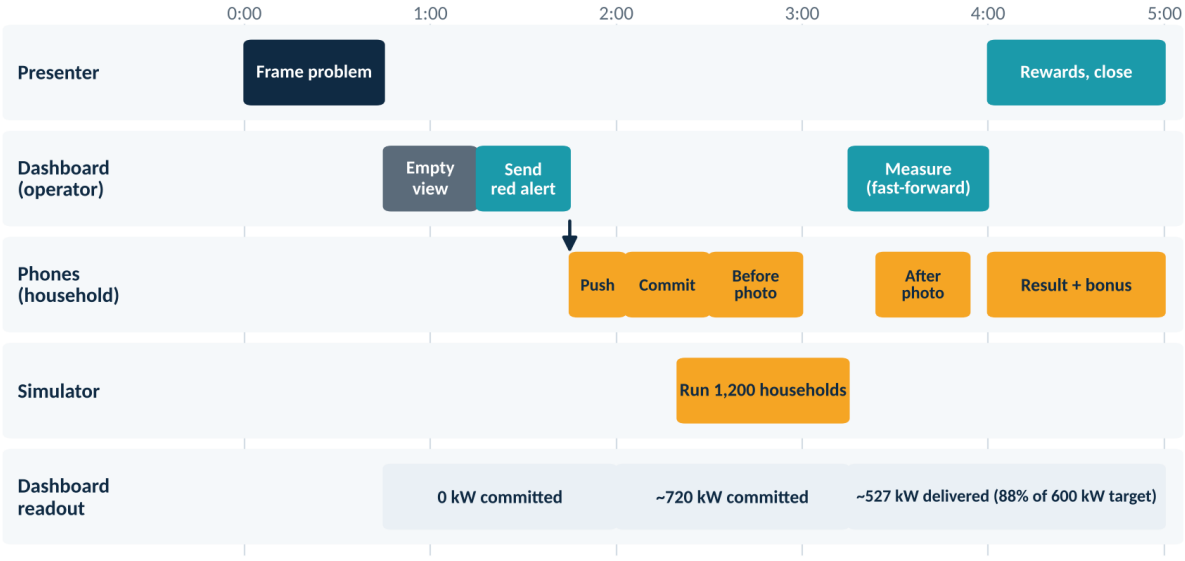
Smart-meter rows in meter_readings come from the AMI feed (simulated in the demo); photo rows come from the app. check_status: pass | flagged | rejected.

7. Demo plan

Story: a red alert hits Luzon. Meralco's operator alerts the Laguna pilot, households commit, and the dashboard shows how much relief arrives. About 5 minutes.

Role	Uses	Job
Presenter	Slides, then the phone	Tells the story and shows the household view
Operator	Web dashboard	Plays Meralco: sends the alert, reads the numbers
Household (2 to 3 people)	Real phones	Receive the push, commit a cut, take before and after photos of the mock meter
Simulator runner	Simulator panel	Runs the 1,200 simulated households
Backup	Laptop with screen recording	Takes over if anything fails

Timeline

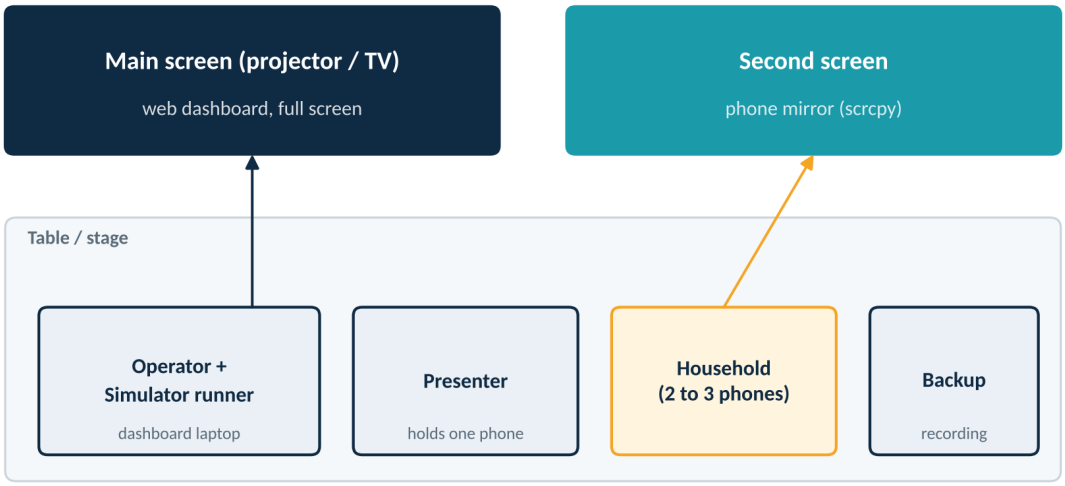


Numbers: 1,200 households x 60% x 1 kW = 720 kW committed. Smart-meter households (15%) at 80% follow-through deliver about 86 kW. Photo households (85%) at 80% follow-through and 90% photo completion deliver about 441 kW. Total about 527 kW, 88% of the 600 kW target. The shortfall is a talking point: every missed photo is verified kW lost, which smart meters would recover. The run is seeded so the presenter's barangay reaches its target and the bonus shows; at 88% overall, most other barangays fall short.

Script

Time	Who	Action	Judges see
0:00	Presenter	Frames the problem with May 14: 12,464 MW supply, 12,877 MW peak.	Slide
0:45	Operator	Shows the empty Laguna dashboard.	Web: 0 committed
1:15	Operator	Red alert, evening 2-hour window, all 7 towns, 600 kW target, preview, confirm.	Web, API, push
1:45	Household	Phones buzz; presenter shows one.	Mobile: alert
2:00	Household	Commit a 1 kW cut.	Mobile; web total ticks up
2:20	Simulator runner	Run simulated households; commits arrive over about 25 seconds.	Web: about 720 kW vs 600 kW target
2:35	Household	Take the before photo of the mock meter and type the reading.	Mobile: photo accepted
3:15	Operator	Measure (fast-forward) to end of window; photo timers are compressed.	Web: about 527 kW delivered, load curve
3:25	Household	Take the after photo; the mock meter now shows the after reading.	Mobile: checks pass
4:00	Presenter	Phone shows the result card with the barangay bonus. Explains that smart-meter households earn the full rate with no photos.	Mobile: result card
4:30	Presenter	Closes: relief at about ILP cost, no generators.	Slide

On-site setup



Everything runs on your own mobile hotspot.

Main screen shows the dashboard. A second screen mirrors one Android phone with scrcpy. Keep two or three real phones on sound and vibrate so judges hear the push arrive.

Bring	Before you go on
USB cable for the mirrored phone, plus a spare.	Connect everything to your own hotspot; check the dashboard loads.
HDMI or USB-C adapter for the projector, and a power strip.	Reset the simulator, then seed. Confirm 0 committed.
Your own mobile hotspot, since venue Wi-Fi may be slow or blocked.	Send one test push to each phone, then reset.

Bring	Before you go on
A screen recording of the full demo.	Do Not Disturb on, with the app allowed. Brightness high; auto-lock off.
A printed one-page script with timings.	Everyone says out loud what they are about to do.
Mock meter: two printed cards with the same serial and the before and after readings, or a tablet showing a meter face.	Register that serial on the demo phones and set their home address to the venue.

If something fails

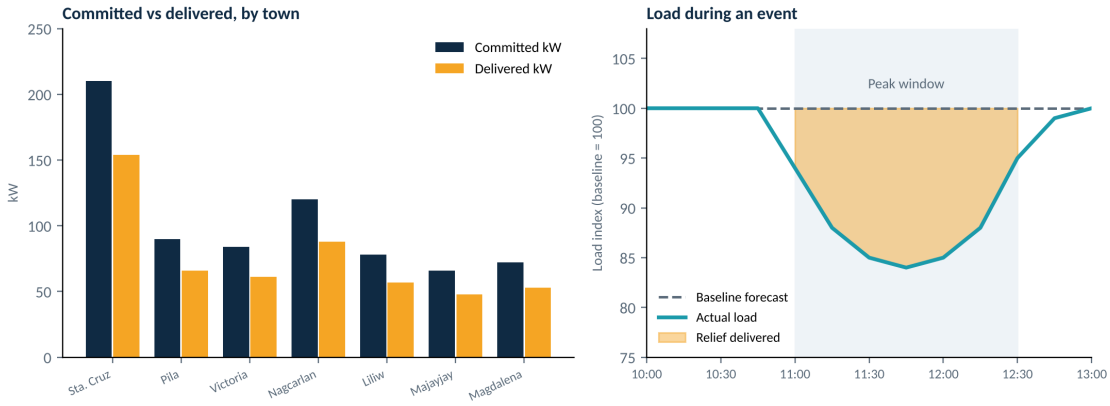
Problem	Response
Phone mirror drops	Hold the phone up for the judges, then reconnect.
Push is slow	Open the app and pull to refresh; the alert is already stored.
Live totals lag	Refresh the dashboard; it polls about every 2 seconds.
Hotspot drops	Switch to the screen recording and talk over it.
Camera or upload fails	Show the result card from a seeded household; the phone flow is on the screen recording.
OCR misreads	OCR only flags and never blocks. Keep it off in the demo environment.
Short on time	Skip the real-phone commit; go from alert straight to the simulator.

Optional second round

Re-run with the poor preset (30%) to show a shortfall and the relief forecast, or target only Santa Cruz barangays to show barangay targeting. Let a judge commit on a phone if time allows.

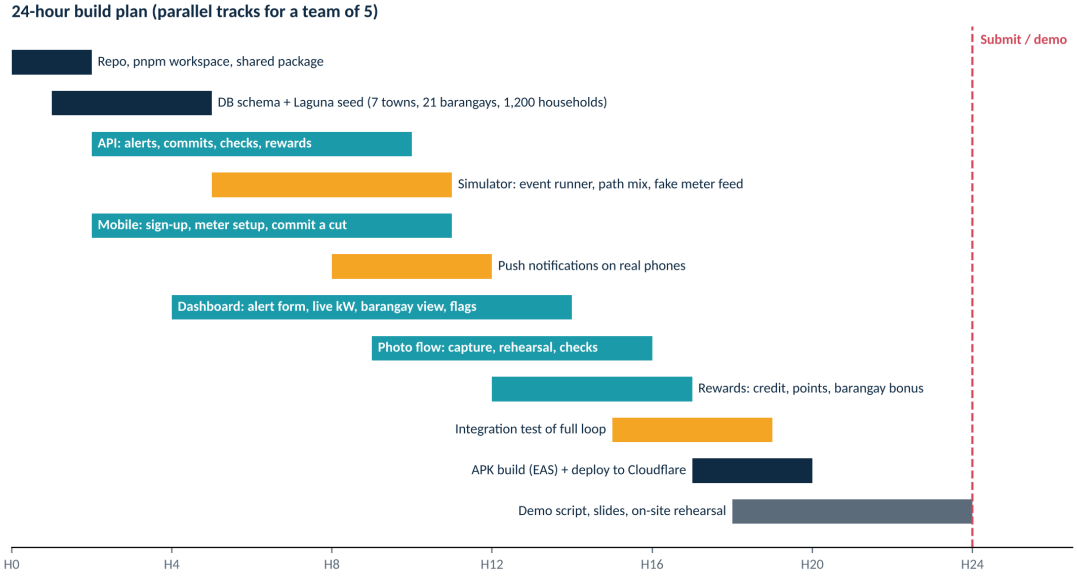
8. Build plan and risks

Dashboard preview (simulated data)



Illustrative layout with simulated numbers: 60% participation, 15% smart-meter households, 90% photo completion.

24-hour build plan



Person	Owns
1	apps/mobile, including the camera and photo flow
2	apps/dashboard
3	apps/api and packages/db, including readings, checks and settlement
4	packages/sim (fake meter feed) and the shared math
5	Pitch, demo script, integration testing

Give each person's AI agent one folder to avoid merge conflicts.

Risks and open decisions

Item	Plan
Verification	Every payout rests on a meter reading: Meralco AMI data, or in-app photos with rule checks and a flagged list. Say plainly in the pitch which parts are simulated.
Smart-meter coverage	About 196,000 smart meters against roughly 8 million customers; Laguna coverage unknown. Plan for most households on the photo path and confirm with Meralco.
Photo burden and drop-off	A rehearsal plus 2 photos per event, with strict windows and no resubmission. Send reminders at window start and end. The 90% completion rate is an assumption to test.
Meter display resolution	Older meters may show whole kWh only, and a 1 kW cut over 2 hours is about 2 kWh. Check what the pilot meters display and prefer longer windows.
Simulated data	Say clearly what is real and what is simulated. Keep simulator parameters visible in the dashboard.
Franchise coverage	Meralco serves only parts of Laguna. Keep the pilot to franchise towns, and confirm Liliw directly with Meralco.
Push and realtime	Test push on real phones by hour 8 to 12. Start with polling for live totals.
Database choice	Neon or Supabase is undecided.
Privacy	Opt-in, results reported per barangay. Meter photos and readings are personal data: private storage, limited retention, stated on the consent screen (Data Privacy Act 2012).
Reward constants	Placeholders: PHP 8.40 per kWh, photo multiplier 0.6, bonus 20%, 100 points per kWh, peak factor 2. Set by Meralco and the ERC in practice.

Background: how the ILP pays

ILP participants are paid the incremental cost of running their own generators: (generator cost per kWh minus grid cost per kWh) × kWh de-loaded, plus maintenance. The 2026 ERC amendment adds a standby fee of PHP 0.85/kWh for capacity committed at final confirmation, and penalties for non-compliance. Cost is recovered from all customers through the generation rate. Sources: ERC Resolution 08 S. 2013, SolarQuarter (July 2026), Meralco ILP presentation (2022), Rappler.

Sources for scope and figures: Philstar (May 14, 2026 red alert), Energy Transition Partnership (DSM case studies), OpenStreetMap franchise list, GMA and Inquirer (Meralco brownout and outage lists). Smart meters: Power Philippines (72,000 smart meters in 2026), BusinessWorld (AMI program), Daily Tribune (full rollout), Insider PH (5-minute interval meters).